

Qualification Pack



Application Development - Android

QP Code: HAR/ITS/Q0601

Version: 1.0

NSQF Level: 4

Haryana State Electronics Development Corporation Limited || SCO 111-113, Sector 17 B
Chandigarh || email:hartron.vikaspanwar@gmail.com

Qualification Pack

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Qualification Pack

HAR/ITS/Q0601: Application Development - Android

Brief Job Description

The qualification impart practical and technical knowledge of Android programming concepts to its learners about building various apps

Personal Attributes

The person should possess problem-solving skills, creativity, and adaptability along with good communication skills

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HAR/ITS/N0601: Computer Basics and Internet](#)
2. [HAR/ITS/N0602: Introduction to Programming with C++](#)
3. [HAR/ITS/N0603: Fundamentals of Java](#)
4. [HAR/ITS/N0605: Querying and Managing data with SQL](#)
5. [HAR/ITS/N0604: Android Programming](#)
6. [HAR/ITS/N0606: Kotlin Fundamentals](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	IT- Services
Occupation	Android Developer
Country	India
NSQF Level	4
Credits	13

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	2512.0204
Minimum Educational Qualification & Experience	12th grade Pass
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	01/11/2026
NSQC Approval Date	01/11/2023
Version	1.0
Reference code on NQR	QG-04-IT-01257-2023-V2-HART
NQR Version	2

Remarks:

Learn Android application development

Qualification Pack

HAR/ITS/N0601: Computer Basics and Internet

Description

Provides the knowledge of fundamental concepts of computers.

Scope

The scope covers the following :

- Understand basics of computer its usage and applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain generation and classification of computer
- PC2.** Identify and explain the various components of computers
- PC3.** Explain operating system, its types and functions.
- PC4.** Define and use internet and Email
- PC5.** Identify common threats and their troubleshooting methods.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The requirement and objective of the organisation and scope of work.
- KU2.** Understanding hardware/software roles and managing files effectively.
- KU3.** Effective use of internet services for various purposes

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Produce work output as per the predefined format and in a timely manner.
- GS2.** Gather feedback from stakeholders and team members to create a user-friendly and impactful design.
- GS3.** Ability to breakdown and solve the problem efficiently.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	-	4	4
PC1. Explain generation and classification of computer	-	-	-	-
PC2. Identify and explain the various components of computers	-	-	-	-
PC3. Explain operating system, its types and functions.	-	-	-	-
PC4. Define and use internet and Email	-	-	-	-
PC5. Identify common threats and their troubleshooting methods.	-	-	-	-
NOS Total	7	-	4	4

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0601
NOS Name	Computer Basics and Internet
Sector	Information Technology Sector
Sub-Sector	
Occupation	Android Developer
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQF Clearance Date	01/11/2023

Qualification Pack

HAR/ITS/N0602: Introduction to Programming with C++

Description

Provide theoretical and practical knowledge of programming with C++

Scope

The scope covers the following :

- Understand various programming techniques, including OOPs concepts in C++

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Define the programming techniques and concepts of OOPs and its applications

PC2. Identify and explain C++ data types, Operators and control structures

PC3. Use various control flow statements to perform task

PC4. Define and use Functions, Arrays and Pointers

PC5. Explain and use classes and Objects, Constructors and Destructors

PC6. Define and use the concepts of inheritance, polymorphism and abstraction

PC7. Differentiate between overloading and overriding

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The organization's goal, requirements, and scope of work.

KU2. Working effectively with team members

KU3. The core programming concept to write codes effectively

KU4. Using exceptions and debugging techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Analyze the requirement of client and provide suitable solution.

GS2. Break down complex problems into smaller, manageable parts.

GS3. Managing time, resources and tasks efficiently to meet deadlines and goals.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	11	10	6	8
PC1. Define the programming techniques and concepts of OOPs and its applications	-	-	-	-
PC2. Identify and explain C++ data types, Operators and control structures	-	-	-	-
PC3. Use various control flow statements to perform task	-	-	-	-
PC4. Define and use Functions, Arrays and Pointers	-	-	-	-
PC5. Explain and use classes and Objects, Constructors and Destructors	-	-	-	-
PC6. Define and use the concepts of inheritance, polymorphism and abstraction	-	-	-	-
PC7. Differentiate between overloading and overriding	-	-	-	-
NOS Total	11	10	6	8

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0602
NOS Name	Introduction to Programming with C++
Sector	Information Technology Sector
Sub-Sector	
Occupation	Android Developer
NSQF Level	4
Credits	2.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQF Clearance Date	01/11/2023

Qualification Pack

HAR/ITS/N0603: Fundamentals of Java

Description

Learn how to develop software using the Java programming language

Scope

The scope covers the following :

- Using Java programming language to write effective codes

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Describe the features of Java programming language

PC2. Define and use basic syntax, semantics & structure of code in Java

PC3. Define various keywords, data types, operators and control flow statements in java.

PC4. Use arrays and Strins in java programming

PC5. Define and use the concept of encapsulation, inheritance, polymorphism and abstraction in java programming.

PC6. Use exceptional handling mechanism to handle errors in java programming.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The organization's requirement, goal and scope of work.

KU2. How to connect and interact with databases.

KU3. How to use algorithms and data structures to solve complex problems.

KU4. How to debug code and identify the root cause of issue.

KU5. How to perform input validation and escaping to ensure security and integrity in the code.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify client needs and deliver the right solution.

GS2. Break down complex problems into smaller tasks.

GS3. Communicate effectively with team members and clients.

GS4. Manage time, resources, and tasks to meet deadlines.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	16	20	10	10
PC1. Describe the features of Java programming language	-	-	-	-
PC2. Define and use basic syntax, semantics & structure of code in Java	-	-	-	-
PC3. Define various keywords, data types, operators and control flow statements in java.	-	-	-	-
PC4. Use arrays and Strings in java programming	-	-	-	-
PC5. Define and use the concept of encapsulation, inheritance, polymorphism and abstraction in java programming.	-	-	-	-
PC6. Use exceptional handling mechanism to handle errors in java programming.	-	-	-	-
NOS Total	16	20	10	10

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0603
NOS Name	Fundamentals of Java
Sector	Information Technology Sector
Sub-Sector	
Occupation	Android Developer
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQF Clearance Date	01/11/2023

Qualification Pack

HAR/ITS/N0605: Querying and Managing data with SQL

Description

provides knowledge of SQL (Structured Query Language) for querying and managing data in relational database.

Scope

The scope covers the following :

- Equips learners with the skills to analyze, query, and manage data in relational database.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define and differentiate database and Relational database management system.
- PC2.** Describe SQL syntax, data types, operators, expression etc.
- PC3.** Explain and use various SQL queries to Insert and manipulate data in database.
- PC4.** Describe and use various keys in database (candidate key, primary key, alternate key and foreign key) and differentiate between them
- PC5.** Explain and use different types of Joins to retrieve data from two or more tables

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Tables, relationships, keys (primary, foreign, and unique), and normalization.
- KU2.** Operations like INSERT, UPDATE, DELETE, and SELECT.
- KU3.** The objective and goal of Organisation and scope of work.
- KU4.** Ensuring efficient and secure handling of organizational data.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Effectively present data in reports and presentations derived from SQL queries.
- GS2.** Collaborate and discuss with teams to define data requirements and solutions.
- GS3.** Structure databases and queries to meet project timelines and organizational needs.
- GS4.** Troubleshoot query performance issues and resolving data inconsistencies.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	11	10	6	6
PC1. Define and differentiate database and Relational database management system.	-	-	-	-
PC2. Describe SQL syntax, data types, operators, expression etc.	-	-	-	-
PC3. Explain and use various SQL queries to Insert and manipulate data in database.	-	-	-	-
PC4. Describe and use various keys in database (candidate key, primary key, alternate key and foreign key) and differentiate between them	-	-	-	-
PC5. Explain and use different types of Joins to retrieve data from two or more tables	-	-	-	-
NOS Total	11	10	6	6

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0605
NOS Name	Querying and Managing data with SQL
Sector	Information Technology Sector
Sub-Sector	
Occupation	Android Developer
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

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HAR/ITS/N0604: Android Programming

Description

Provide knowledge of Android Programming

Scope

The scope covers the following :

- To design, test, and develop application using Android platform

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define Android and its Architecture
- PC2.** Explain the procedure steps of Installing Android SDK Tools
- PC3.** Explain the core building blocks of Android and develop a simple application.
- PC4.** Define Activity and the life cycle methods of android activity.
- PC5.** Use Android widgets, intents, themes, menus, and services.
- PC6.** Integrate multimedia, camera and Location based services.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The requirement and objective of the organisation and scope of work.
- KU2.** Fundamental understanding of programming paradigms
- KU3.** Development Framework and Handling of Data.
- KU4.** Writing clean, maintainable code and Basic debugging, testing, and performance optimization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to think critically to identify problems and implement solutions quickly.
- GS2.** Effectively manages time and resources to ensure timely completion of tasks.
- GS3.** Adapts to new tools and techniques to address evolving project requirements.
- GS4.** Work with team members, guiding them and providing support to junior developers

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	-	8	4
PC1. Define Android and its Architecture	-	-	-	-
PC2. Explain the procedure steps of Installing Android SDK Tools	-	-	-	-
PC3. Explain the core building blocks of Android and develop a simple application.	-	-	-	-
PC4. Define Activity and the life cycle methods of android activity.	-	-	-	-
PC5. Use Android widgets, intents, themes, menus, and services.	-	-	-	-
PC6. Integrate multimedia, camera and Location based services.	-	-	-	-
NOS Total	15	-	8	4

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0604
NOS Name	Android Programming
Sector	Information Technology Sector
Sub-Sector	
Occupation	Android Developer
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

HAR/ITS/N0606: Kotlin Fundamentals

Description

Provides introduction to Kotlin language used for Android development.

Scope

The scope covers the following :

- Provides the foundational concepts of Kotlin, which can be applied to develop Android applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Define Kotlin and its features.

PC2. Define syntax, variables and data types in Kotlin

PC3. Define syntax, variables and data types in Kotlin

PC4. Explain and use operators and control flow structures to create programs in Kotlin

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Kotlin's syntax, including variables, data types, and expressions.

KU2. Kotlin's application in Android mobile app development.

KU3. The Organisational objective and goals and scope of work.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Work efficiently with team members.

GS2. Use basic control structures (if, for loops) based on the problem requirements.

GS3. Design simple and efficient programs that meet the needs of the end-users.

GS4. Effectively prioritize coding tasks based on deadlines, project requirements, and complexity.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	-	4	4
PC1. Define Kotlin and its features.	-	-	-	-
PC2. Define syntax, variables and data types in Kotlin	-	-	-	-
PC3. Define syntax, variables and data types in Kotlin	-	-	-	-
PC4. Explain and use operators and control flow structures to create programs in Kotlin	-	-	-	-
NOS Total	9	-	4	4

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0606
NOS Name	Kotlin Fundamentals
Sector	Information Technology Sector
Sub-Sector	
Occupation	Android Developer
NSQF Level	4
Credits	1.5
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment is done online after completion of the course. Assessment comprises objective, subjective and practical questions

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N0601.Computer Basics and Internet	7	-	4	4	15	7
HAR/ITS/N0602.Introduction to Programming with C++	11	10	6	8	35	18
HAR/ITS/N0603.Fundamentals of Java	16	20	10	10	56	28
HAR/ITS/N0605.Querying and Managing data with SQL	11	10	6	6	33	17
HAR/ITS/N0604.Android Programming	15	-	8	4	27	14
HAR/ITS/N0606.Kotlin Fundamentals	9	-	4	4	17	8
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	8
Total	89	70	38	36	233	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.